

US Accidents Project

Final Review — All 8 Dashboards Confirmed

This is the final review document based on the actual completed Power BI file (Final.pdf). All 8 dashboards have been verified against the real screenshots. This document summarises the confirmed figures, the one remaining inconsistency found, and the corrected Executive Summary for Dashboard 6.

Section 1 — All 8 Dashboards: Confirmed Status

#	Dashboard Name	Key Numbers Confirmed	Status
1	Overview	500K accidents Avg Severity 2.50 High Severity 19.52% Avg Duration 421.9 min	COMPLETE
2	Temporal Analysis	202K rush hour 80K weekend Friday busiest Dec peak (54,589)	COMPLETE
3	Weather & Environment	26K extreme 37K rain 10K snow 12K fog 94.75% normal weather	COMPLETE
4	Geography	CA 113K FL 57K TX 37K Miami top city I-95 N top street	COMPLETE
5	Road Features	74K signal 37K junction 56K crossing 351K with 0 road features	COMPLETE
6	ML Model Results	Best ROC-AUC: CatBoost 74.60% Best Accuracy (Binary): LightGBM 78.62%	REVIEW NOTE
7	Forecast (Prophet)	497K historical 124K predicted Dec peak 14,415 Jun lowest 6,731	COMPLETE
8	ML Predictions (CatBoost)	363 High Risk (36.3%) 637 Low Risk (63.7%) Avg Score 0.44 Threshold 0.5034	COMPLETE

Section 2 — Confirmed Key Figures Per Dashboard

Dashboard 1 — Overview KPIs

500K Total Accidents	2.50 Avg Severity	421.9 Avg Duration (min)	19.52% High Severity %
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Severity breakdown: 79.63% Moderate Impact | 16.9% High Impact | 2.61% Critical Impact | ~0.87% Low Impact

Top states by total accidents: CA, FL, TX, SC, NY, NC, VA, PA, MN, OR

Dashboard 2 — Temporal Analysis

202K Rush Hour Accidents	80K Weekend Accidents
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Metric	Value	Notes
Busiest Day	Friday	Highest total accidents across all years
Quietest Day	Saturday	Significantly lower than weekdays
Peak Month	December	54,589 total across all years
Lowest Month	March	35,557 total across all years
Peak Year	2022	113,734 accidents — highest single year
Rush Hour Pattern	7–9 AM & 15–18	Two clear daily peaks visible in hourly chart

Dashboard 3 — Weather & Environment

26K Extreme Weather	37K Rain Accidents	10K Snow Accidents	12K Fog Accidents
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Weather Condition	Total Accidents	Rank
Fair	177K	1st — majority of accidents in clear weather
Mostly Cloudy	65K	2nd
Cloudy	53K	3rd
Clear	52K	4th
Partly Cloudy	45K	5th
Overcast	25K	6th
Light Rain	23K	7th
Scattered Clouds	13K	8th
Light Snow	8K	9th

Dashboard 4 — Geography

Category	Top Entry	Value	2nd	3rd
State	California	113K	Florida 57K	Texas 37K
City	Miami	~14K	Houston	Los Angeles
Street	I-95 N	~5K	I-95 S	I-5 N

Dashboard 5 — Road Features

74K Signal Accidents	37K Junction Accidents	56K Crossing Accidents	0.42 Avg Road Features
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Road Feature	Critical	High Impact	Low Impact	Moderate	Key Insight
Crossing	816	3,185	1,171	51,081	Highest moderate volume

Road Feature	Critical	High Impact	Low Impact	Moderate	Key Insight
Junction	1,324	8,693	208	26,782	High critical + high impact rate
Station	170	819	174	11,835	Consistent across severities
Amenity	132	276	87	5,696	Lower volume feature
Railway	93	566	69	3,592	Moderate risk feature

Signal present: 14.81% of accidents | At junction: 7.4% of accidents | 70.2% of accidents occur with 0 road features present.

Section 3 — Dashboard 6: ML Results — Review Note

Dashboard 6 has been updated correctly and shows CatBoost V4 as the top ROC-AUC model. However there is one remaining inconsistency to be aware of:

3.1 The Accuracy Card Shows 78.62% — This Needs Clarification

The large KPI card at the top of Dashboard 6 displays 78.62% Accuracy. This is the Binary Accuracy of LightGBM, not CatBoost. This is because the card is calculating MAX(Accuracy) across all models. This is misleading — a visitor will think 78.62% is the accuracy of your production model.

Model	Task	Accuracy	ROC-AUC	Production?
LightGBM	Binary	78.62%	50.00%	No — high accuracy but ROC-AUC = random
XGBoost	Binary	76.85%	50.00%	No — same problem
CatBoost V4 (Production)	Binary	69.76%	74.60%	YES — best real-world performance
Random Forest	Binary	68.10%	50.10%	No
Logistic Regression	Binary	50.06%	49.90%	No
LightGBM	Multi	69.83%	—	No
Random Forest	Multi	66.41%	—	No
XGBoost	Multi	68.34%	—	No
Logistic Regression	Multi	43.94%	—	No

Why LightGBM and XGBoost have high accuracy but low ROC-AUC:

- A ROC-AUC of ~0.50 means the model is performing at random — it cannot distinguish between High Risk and Low Risk cases.
- LightGBM Binary 78.62% accuracy is likely achieved by predicting the majority class almost always — not by learning real patterns.
- CatBoost V4 with ROC-AUC = 0.746 genuinely separates the two risk classes, which is why it was chosen as the production model.
- The optimised threshold (0.5034) further improves the balance between precision and recall.

ACTION — Fix the Accuracy KPI Card in Dashboard 6:

What to Change	Current Value	Fix To
Accuracy KPI card	MAX(Accuracy) = 78.62%	Filter card to CatBoost V4 only: CALCULATE(MAX(ML_Results[Accuracy]), ML_Results[Model] = "CatBoost V4 (Production)") — shows 69.76%
Card label	Accuracy	Best Model Accuracy (CatBoost V4)
Executive Summary	LightGBM 69.83% Multiclass mentioned as primary model	Update to: Best Production Model = CatBoost V4, ROC-AUC 74.60%, Accuracy 69.76% (see Section 3.2)

3.2 Corrected Executive Summary Text Box

Replace the current text in the Dashboard 6 Project Executive Summary box with:

Architecture: Medallion Pipeline (Bronze > Silver > Gold) — Python ETL + Spark/SQL refinement.

Data Volume: 500,000 US accident records (2016–2023) — fully cleaned and modelled.

Best Model: CatBoost V4 (Binary Classification) — ROC-AUC: **0.746**, Accuracy: 69.76%, Optimised Threshold: 0.5034.

Why CatBoost: LightGBM achieves 78.62% accuracy but ROC-AUC = 0.50 (random). CatBoost ROC-AUC of 0.746 confirms genuine class separation.

Top Features: Distance(mi), Wind_Speed_mph, Year, Temperature_F, Pressure_in, Week_of_Year, Humidity_pct.

Forecast: Prophet model (87 months training) predicts 124K accidents in the next 12 months. December is the peak risk month.

Section 4 — Dashboards 7 & 8: Confirmed Final Numbers

Dashboard 7 — Forecast (Prophet Model)

497K Historical Accidents	124K Predicted Next 12M	14K Peak Month (Dec)	7K Lowest Month (Jun)
Metric	Value	Notes	
Peak predicted month	December	14,415 accidents — highest risk month	
Lowest predicted month	June	6,731 accidents — lowest risk month	
Winter vs Summer	+40%	Winter months (Dec–Jan) vs summer average	
Training data	87 months	2016–2023 historical monthly data	
Confidence interval	95%	Upper and lower bounds shown in area chart	
Forecast horizon	12 months	2024 full-year monthly predictions	

Dashboard 8 — ML Predictions (CatBoost V4)

363 High Risk (3-4)	637 Low Risk (1-2)	0.44 Avg Risk Score	36.30% High Risk %
Metric	Value	Notes	
Total records scored	1,000	Random sample from Gold layer dataset	
High Risk classified	363	36.3% — probability ≥ 0.5034	
Low Risk classified	637	63.7% — probability < 0.5034	
Average risk score	0.44	Below threshold — distribution skewed toward low risk	
Decision threshold	0.5034	Optimised threshold — confirmed in histogram visual	
Top risk weather code	42	Weather_Condition_Encoded 42 — avg score 0.79 (2016)	
Scatter chart	Confirmed	X=Temperature_F Y=Risk_Score_Prob Color=Risk Level	
Model info bar	Confirmed	CatBoost V4 Threshold 0.5034 Features 20 ROC-AUC 0.746	

Both dashboards are confirmed complete and match the guidance in the previous review document. No structural changes needed. Only the Dashboard 6 Accuracy card fix (Section 3) remains.

Section 5 — Complete Project Summary

5.1 Full 8-Dashboard Overview

Dashboard	Story It Tells	Most Important Number
1 Overview	Scale and severity of US accidents	500K records 19.52% high severity
2 Temporal	When accidents happen most	Friday peak 202K rush hour Dec 54,589
3 Weather	Environmental conditions during accidents	177K in Fair weather — not just bad weather
4 Geography	Where accidents concentrate	CA 113K Miami I-95 top corridor
5 Road Features	Infrastructure impact on severity	70% with 0 features Junction = more severe
6 ML Results	Model comparison and feature importance	CatBoost ROC-AUC 0.746 — best classifier
7 Forecast	Predicted future accident trends	124K next 12 months Dec highest risk
8 Predictions	Live model scoring on sample records	36.3% High Risk Threshold 0.5034 confirmed

5.2 One Remaining Action Item

Item	Location	Action Required
Accuracy KPI card shows 78.62% (LightGBM) instead of CatBoost figure	Dashboard 6 — top left KPI card	Change DAX to filter for CatBoost V4 only. Result will show 69.76%. Rename card label to: Best Model Accuracy (CatBoost V4)

5.3 Presentation Order (Recommended)

Order	Dashboard	Opening Line for Presentation
1	Overview	'We analysed 500,000 US road accidents from 2016 to 2023.'
2	Geography	'California alone accounts for 113K — nearly a quarter of all accidents.'
3	Temporal	'Friday rush hour is the single most dangerous time window.'
4	Weather	'Surprisingly, most accidents happen in fair weather — not storms.'
5	Road Features	'Junctions and crossings significantly increase accident severity.'
6	ML Results	'We tested 5 models — CatBoost showed the strongest real-world performance.'
7	Forecast	'Prophet predicts 124K accidents in the next 12 months — December peaks.'
8	ML Predictions	'Here is our model scoring live data — 36% flagged as High Risk.'